

AOT-GOAT: dataset report

Which data was used, where it came from, what was done with it, and how much of it is real. Figures are from the final runs in the delivered ZIP (aot-goat.zip).

Summary

One real dataset was used, as an unlabeled robustness check only. Every accuracy figure comes from synthetic or hand-written data. No human-annotated data and no labeled real conversations were used.

	Real	Synthetic / hand-written
Labeled evaluation cases	0	2,579 (59 hand-written + 720 calibration + 1,800 test)
Unlabeled messages processed	2,000 real dialogue turns	-
Basis for accuracy claims	None	All of it
Human-annotated labels	None	None

1. Real data (1 dataset)

Item	Detail
Dataset	MultiWOZ 2.2, one shard (dialogues_001.json): human-written task-oriented travel-booking dialogues (512 dialogues in the file).
Source	github.com/budzianowski/multiwoz . Licence listed as MIT; verify before citing. Papers: Budzianowski et al. 2018 (MultiWOZ); Zang et al. 2020 (v2.2).
Obtained how	Downloaded on demand by <code>python evaluate.py --real</code> . Not included in the ZIP.
Sample used	2,000 system turns, randomly sampled with a fixed seed, each with up to 8 earlier turns as history.
Labels	None. System turns were treated as messages to the user.

What was done: the pipeline was run on each turn with and without its history. Sender context was built automatically (a pronoun's referent = last noun phrase of the previous turn).

What it showed:

- 0 crashes and 0 constraint violations across the adapted outputs.
- Real history lowered flag rates: coreference 21.2% to 13.6%, temporal 8.6% to 5.5%, scope 20.1% to 16.7%.
- It exposed one checker false alarm (messages with no final punctuation), which was fixed.
- **Not used for** training, tuning, threshold calibration or accuracy measurement. A trial of mining its clarification questions as weak labels was dropped: the patterns were too noisy.

Enron, Ubuntu IRC, ClarQ and CLAUQA were suggested as options earlier but were **not** downloaded or used.

2. Synthetic and hand-written data

Dataset	Cases	Made by	Used for
Demo scenarios (data/sample_data.json)	2 messages, 5 recipient histories	Written by the AI assistant	UI demo
Controlled set, dev split	41	Hand-written	Building and debugging the rules
Controlled set, holdout split	18	Hand-written, after the rules were done	Checking only, never tuned on
Generated calibration set (seed 1)	720 (120 simulated dyads)	Seeded generator	Choosing the threshold (0.35); finding bugs
Generated test set (seed 2)	1,800 (300 simulated dyads)	Seeded generator	Reported results, scored once with the threshold frozen

How the generator labels cases. Each simulated recipient has a history, a time zone and known terms. Gold labels come from how a case was built (was the risky span grounded in the history or not), never from the pipeline's output. Test set slices:

Slice	Cases	What it contains
single	900	One risk type per message, grounded or ungrounded
mixed	300	2-3 risk types in one message, each independently grounded or not

Slice	Cases	What it contains
hard_pos	300	Risky wording outside the rule vocabulary (e.g. 'what we did for Mumbai', lowercase acronyms)
hard_neg	300	Idioms that look risky ('Welcome to the team!', 'Everything is all set.')

Gold-positive rate on the test set: 12.6% of (case, risk type) pairs. Development debugging also used extra runs of seed 1 (not reported).

3. What the data showed (test seed 2)

Setting	Precision	Recall	F1 (95% CI)
Pipeline, designed scenarios (single + mixed, 1,200 cases)	1.000	0.998	0.999 (0.997-1.000)
Same, recipient profile switched off	0.571	1.000	0.727 (0.707-0.747)
Pipeline, all 1,800 cases incl. hard slices	0.830	0.774	0.801 (0.787-0.816)
Same, recipient profile switched off	0.522	0.776	0.624 (0.604-0.645)

- Using the recipient's history adds 0.25 to 0.29 F1 over the identical detector without it (95% CI, resampling dyads).
- hard_pos recall is 0.153. On hard_neg every flag is a false positive (104 of 1,500 type decisions).
- Controlled 59-case set: dev F1 0.963, holdout F1 0.963.
- 610 adapted outputs: all passed intent, no-invented-facts and minimal-edit checks; an independent check against generator facts found 0 untraceable tokens and 0 polarity changes.

4. Limitations and what to claim

- The generator and the rules share an author and a view of what 'grounded' means. The near-perfect designed-scenario score shows consistency and the benefit of the profile, not real-world accuracy.
- Bugs found on the calibration seed were fixed before the test seed was scored. One checker false alarm found later by the real-text test was fixed afterwards; it affects safety checks only.
- **Safe to say:** on seeded synthetic dyads with a frozen threshold, recipient conditioning improved F1 by 0.25-0.29 over the same detector without it; weak recall on unseen wording is reported; 2,000 real dialogue turns ran without crashes or constraint violations.
- **Not safe to say:** any accuracy on real conversations, or validation on a real labeled dataset.
- **Missing step:** independent human annotation of real conversations and human ratings of 'same intent / no new facts'.